Final Report: Fashion Outfit Compatibility

Yayun Tan
1009863756, yayun.tan@mail.utoronto.ca
github.com/YayunTan/aps360_outfit_completion

1 Project Description

Choosing an outfit is a common but often frustrating daily decision. Although people may own many clothing items, they still struggle to combine them into outfits that feel visually compatible and appropriate. In this project, I study image-based clothing compatibility ranking using the DeepFashion dataset. The task is formulated as follows: given a query clothing image, the model ranks candidate items from allowed target categories by compatibility. The goal is to retrieve the Top-5 most compatible items for each query image. A CNN-based model can therefore be used to learn visual representations that support compatibility ranking.

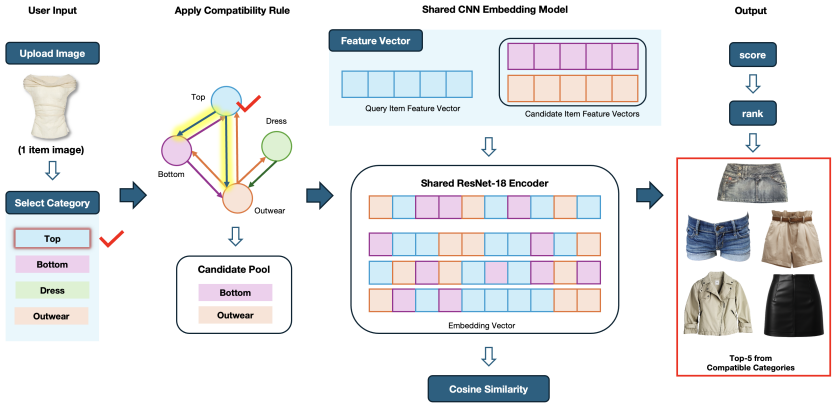

Figure 1: Model architecture illustration.

2 Background & Related Works

(1) Outfit Compatibility Prediction and Diagnosis with Multi-Layered Comparison Network. Wang et al. propose the Multi-Layered Comparison Network (MCN) for predicting outfit compatibility and diagnosing incompatibility factors. Their work shows that compatibility depends on multi-level visual cues, such as color, texture, and style. [1]

(2) OutfitTransformer: Learning Outfit Representations for Fashion Recommendation. OutfitTransformer uses self-attention to learn outfit-level representations from sets of items. Although my model is not transformer-based, this work highlights the importance of modeling relationships among multiple outfit items. [2]

(3) Dressing as a Whole: Outfit Compatibility Learning Based on Node-wise Graph Neural Networks. Cui et al. model outfits as graphs and learn compatibility through message passing. This shows the value of capturing interactions between items rather than treating them independently. [3]

(4) Fashion Outfit Complementary Item Retrieval. Lin et al. study complementary item retrieval, which is closely related to my task of ranking candidate items and returning the Top-5 most compatible results. [4]

(5) DeepFashion: Powering Robust Clothes Recognition and Retrieval With Rich Annotations. DeepFashion is a large-scale benchmark for clothing recognition and retrieval with rich annotations, making it a standard dataset for fashion understanding tasks. [5]

3 Data Processing

I use the DeepFashion Category and Attribute Prediction Benchmark, which contains clothing images together with category labels, attribute annotations, and predefined train/validation/test partitions. To simplify the original label space and make the task easier to define consistently, I group the fine-grained categories into 4 coarse categories: top, bottom, dress, and outerwear. All remaining categories are discarded for the current version of the project. I then define rule-based compatibility relations between the coarse categories. Before being passed into the model, all images are resized to 224×224 and normalized using standard ImageNet mean and standard deviation so that they are compatible with the pretrained ResNet backbone.

In addition to category annotations, I also use the DeepFashion attribute annotations. Each image is associated with a 1000-dimensional attribute vector. After binarizing the attributes, I use attribute overlap as a proxy for compatibility, which provides a more meaningful basis for evaluation than random candidate assignment.

```
allowed_pairs = {
    "top": {"bottom", "outerwear"},
    "bottom": {"top", "outerwear"},
    "dress": {"outerwear"},
    "outerwear": {"top", "bottom", "dress"},
}
```

coarse_category	
top	128000
dress	88430
bottom	58963
outerwear	13829

split	
train	209222
val	40000
test	40000

Figure 2: Rule-based compatibility mapping, category distribution, and dataset split counts.

4 Baseline Model

The initial baseline is a Frozen CNN + cosine similarity retrieval model. Each observed item image is encoded by a frozen pretrained CNN backbone (ResNet-18) into an embedding vector. For a query item, the model retrieves candidates only from rule-allowed target categories and ranks them by cosine similarity, returning the Top-5 suggestions. To define supervision, the candidate with the highest attribute overlap is treated as the pseudo-positive, while low-overlap candidates are used as negatives. The baseline is evaluated using Hit@1, Hit@5, and Mean Reciprocal Rank (MRR). Hit@K measures whether the positive item appears within the top-K results, while MRR also rewards higher ranking positions. For efficiency, evaluation is conducted on a balanced subset of 4,000 validation queries and 4,000 test queries, with 1,000 queries sampled from each coarse category.

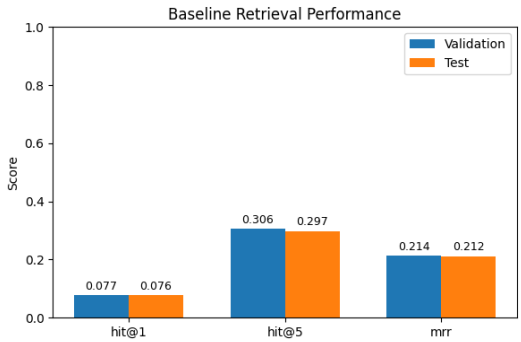

Figure 3: Baseline model results.

Overall, the baseline shows that image-based retrieval is feasible, but performance is still limited because the model relies only on frozen visual similarity and does not explicitly learn compatibility.

5 Primary Model

The primary model is a fine-tuned ResNet-18 ranking model trained with triplet loss. Unlike the baseline, which uses frozen pretrained features, this model fine-tunes the image encoder so it can learn features more relevant to fashion compatibility. The ResNet-18 backbone extracts visual features, which are then passed through a projection head into a lower-dimensional embedding space and normalized for similarity comparison. The same encoder is shared across the anchor, positive, and negative images, allowing the model to learn a unified compatibility-aware embedding space. During inference, the query and candidate images are encoded by the same network, and candidates are ranked by similarity to return the Top-5 suggestions.

The model is trained with triplet loss, where each example contains an anchor item, a more compatible positive item, and a less compatible negative item. The loss encourages the anchor to be closer to the positive than to the negative by a fixed margin. In the main setting, the model uses an embedding dimension of 128, the Adam optimizer, a batch size of 32, and 5 training epochs. Due to computational limits, training was conducted on 4,000 samples per category rather than the full dataset.

5.1 Quantitative Results

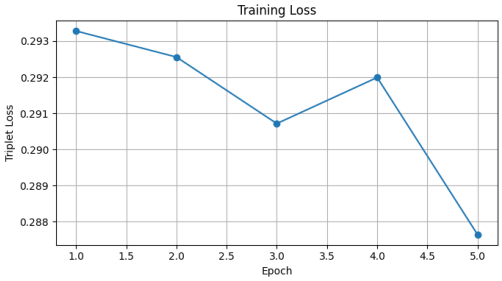

Figure 4: Training loss of the primary mode.

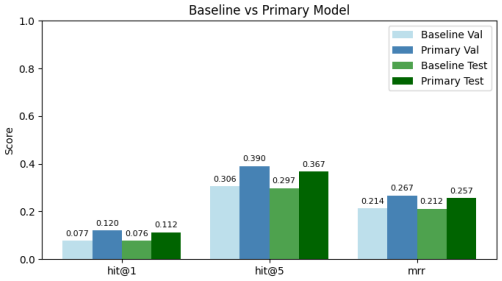

Figure 5: Comparison of baseline and primary model performance.

The training loss curve (Fig. 4) shows a downward trend over the five epochs, decreasing from about 0.2933 to 0.2876, which suggests that the model was learning to separate positive and negative samples more effectively in the embedding space.

According to the results showing in Figure 5, the primary model outperformed the baseline on all three ranking metrics for both the validation and test sets. On the validation set, Hit@1 improved from 0.077 to 0.120 (+55.8%), Hit@5 improved from 0.306 to 0.390 (+27.5%), and MRR improved from 0.214 to 0.267 (+24.8%). On the test set, Hit@1 increased from 0.076 to 0.112 (+47.4%), Hit@5 increased from 0.297 to 0.367 (+23.6%), and MRR increased from 0.212 to 0.257 (+21.2%). These results suggest that fine-tuning the ResNet-18 encoder and training with triplet loss helped the model learn a more compatibility-aware embedding space than the frozen-feature baseline.

5.2 Qualitative Results

Figure 6: Sample query with a white t-shirt top.

For qualitative evaluation, I focused on both category compatibility and style compatibility. While category compatibility checks whether the retrieved item comes from an allowed target category,

style compatibility is more important because it reflects whether the model retrieves items that are visually coherent with the query beyond simple rule-based filtering.

Figure 6 showed the results of a sample query. For this query top, the model retrieves outerwear and bottom items, which are consistent with the rule-based compatibility constraints. The retrieved results here are more visually coherent: most outputs share light color tones, simple silhouettes, and layering styles that plausibly complement the query dress.

However, in some weaker examples, the model could fail to return stylistically coherent matches and pair a query skirt with items resembling men's formalwear. This suggests that category correctness alone is not sufficient, and that style matching is a more informative qualitative criterion for this task.

6 Evaluation on New Data

To evaluate generalization beyond the DeepFashion benchmark, we tested the final model on a small unseen set of retail-style clothing images downloaded from Zara and H&M (20 queries, 100 candidates).

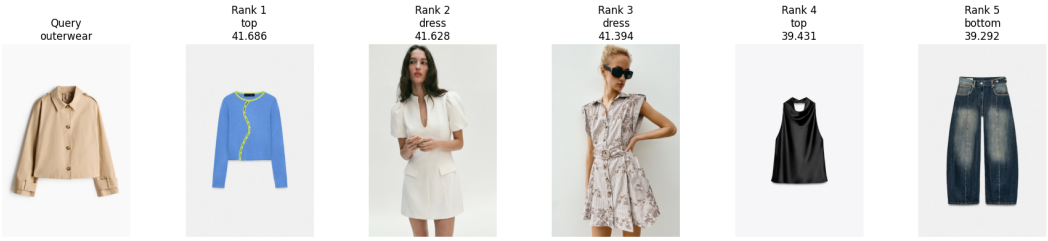

Figure 7: Sample query with new image collections.

As shown in Fig. 7, the model retrieved items from rule-compatible target categories for a query outerwear item, including tops, dresses, and bottoms. Although the similarity scores on this new retail data were generally lower than those observed on the DeepFashion validation and test sets, the retrieved results were still visually reasonable and stylistically coherent. In particular, the returned items shared a clean, minimal, and neutral-toned aesthetic that plausibly matched the query outerwear. This suggests that the model learned some transferable compatibility cues, even though its performance on new data was weaker than on the benchmark dataset.

7 Discussion

The primary model outperformed the baseline on all ranking metrics, showing that fine-tuning with triplet loss helped learn a more compatibility-aware embedding space. Qualitative examples suggest that the model often retrieves more visually coherent matches than the baseline, especially for simple items with clear styling cues. However, the results also show that category correctness alone is not sufficient: in weaker cases, the model returned stylistically inappropriate matches even when the rule-based category constraints were satisfied. Overall, the model learned useful compatibility patterns, but its ability to generalize to unseen retail data remains limited.

8 Ethical Consideration

There is a risk that automated recommendations could homogenize personal style by steering users towards algorithmically preferred combinations. My model should be positioned as a tool for inspiration rather than prescription. While DeepFashion data is public, in a real-world deployment, users' wardrobes and preferences would be sensitive personal information that requires careful privacy protections. Additionally, fashion items in DeepFashion are shown on models or product images that may not represent diverse body types. My model does not account for how items fit different bodies, and it could lead to recommendations that work aesthetically but not practically for all users.

References

[1] X. Wang, B. Wu, Y. Ye, and Y. Zhong, "Outfit compatibility prediction and diagnosis with multi-layered comparison network," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 329–337, doi: 10.1145/3343031.3350909.

[2] R. Sarkar *et al.*, "OutfitTransformer: Learning outfit representations for fashion recommendation," arXiv:2204.04812, Apr. 15, 2022, doi: 10.48550/arXiv.2204.04812.

[3] Z. Cui, Z. Li, S. Wu, X. Zhang, and L. Wang, "Dressing as a whole: Outfit compatibility learning based on node-wise graph neural networks," in *Proc. World Wide Web Conf.*, May 2019, pp. 307–317, doi: 10.1145/3308558.3313444.

[4] Y.-L. Lin, S. Tran, and L. S. Davis, "Fashion outfit complementary item retrieval," arXiv:1912.08967, Mar. 4, 2020, doi: 10.48550/arXiv.1912.08967.

[5] Z. Liu, P. Luo, S. Qiu, X. Wang, and X. Tang, "DeepFashion: Powering robust clothes recognition and retrieval with rich annotations," in *2016 IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 1096–1104, doi: 10.1109/CVPR.2016.124.